

Low Dimension Lie Algebras

1.1 Dimension 3

1.1.1 Lie Algebras where $L' = L$

Theorem 1.1.1

Suppose that L is a complex Lie algebra of dimension 3 such that $L = L'$. Then L is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proof. 1. Let $x \in L$ be non-zero. We claim that $\text{ad } x$ has rank 2. Extend x to a basis of L say $\{x, y, z\}$. Then L' is spanned by

$$\{[x, y], [x, z], [y, z]\}$$

But $L' = L$, this set must be linearly independent, and hence the image of $\text{ad } x$ has a basis $\{[x, y], [x, z]\}$ of size 2, as required.

2. We claim that there is some $h \in L$ such that $\text{ad } h$ has an eigenvector with a non-zero eigenvalue. Choose any non-zero $x \in L$. If $\text{ad } x$ has a non-zero eigenvalue, then we take $h = x$. If $\text{ad } x$ has no non-zero eigenvalues, then, as it has rank 2, its Jordan canonical form must be

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

This matrix indicates there is a basis of L extending $\{x\}$, say $\{x, y, z\}$ such that $[x, y] = x$ and $[x, z] = y$. So $\text{ad } y$ has x as an eigenvector with eigenvalue -1 , and we may take $h = y$.

3. By the previous step, we may find $h, x \in L$ such that $[h, x] = \alpha x \neq 0$. Since $h \in L$ and $L = L'$, we know $\text{ad } h$ has trace zero. It follows that $\text{ad } h$ must have three distinct eigenvalues $\alpha, 0, -\alpha$. If y is an eigenvector for $\text{ad } h$ with eigenvalue $-\alpha$, then $\{h, x, y\}$ is a basis of L . In this basis, $\text{ad } h$ is represented by a diagonal matrix.

4. To determine $[x, y]$. Note that

$$[h, [x, y]] = [[h, x], y] + [x, [h, y]] = \alpha[x, y] + (-\alpha)[x, y] = 0$$

5. We now make two applications of step 1. Firstly, $\ker \operatorname{ad} h = \operatorname{span} \{h\}$, so $[x, y] = \lambda h$. Secondly, $\lambda \neq 0$ otherwise $\ker \operatorname{ad} x$ is 2-dimensional. By replacing x with $\lambda^{-1}x$ we may assume that $\lambda = 1$.

□

Proposition 1.1.2

Suppose that $\alpha, \beta \in \Phi$ and $\beta \neq \pm\alpha$.

1. $\beta(h_\alpha) \in \mathbb{Z}$.
2. There are integers $r, q \geq 0$ such that $\beta + k\alpha \in \Phi$ if and only if $k \in \mathbb{Z}$ and $-r \leq k \leq q$. Moreover, $r - q = \beta(h_\alpha)$.
3. If $\alpha + \beta \in \Phi$, then $[e_\alpha, e_\beta]$ is a non-zero scalar multiple of $e_{\alpha+\beta}$.
4. $\beta - \beta(h_\alpha)\alpha \in \Phi$.

Proof. Let $M := \bigoplus_k L_{\beta+k\alpha}$ be the root string of α through β .

1. $\beta(h_\alpha)$ is the eigenvalue of h_α acting on L_β , so it lies in $\mathbb{Z}$.
2. $\dim L_{\beta+k\alpha} = 1$ whenever $\beta + k\alpha$ is a root, so the eigenspaces of $\operatorname{ad} h_\alpha$ on M are all 1-dimensional and, since $(\beta + k\alpha)h_\alpha = \beta(h_\alpha) + 2k$, the eigenvalues of $\operatorname{ad} h_\alpha$ on M are either all even or all odd.

□

1.2 Cartan Subalgebras as Inner-Product Spaces